

where $h = H_0/M$ and $\beta = H_a/M$ are dimensionless magnetic field and anisotropy constant, respectively; and the dipolar sums $\sigma(\vec{k})$ and $\sigma_c(\vec{k})$ appear

$$\begin{aligned}\sigma(\vec{k}) &= \sum_{\vec{l} \neq 0} \frac{1}{(l_x^2 + l_y^2)^{3/2}} e^{i\vec{k}\vec{l}}, \\ \sigma_c(\vec{k}) &= \sigma'(\vec{k}) + i\sigma''(\vec{k}) = \sum_{\vec{l} \neq 0} \frac{(l_x - il_y)^2}{(l_x^2 + l_y^2)^{3/2}} \cdot e^{i\vec{k}\vec{l}}.\end{aligned}\quad (7)$$

Such sums naturally emerge for any problem involving interaction of dipoles ordered in a lattice. For the two-dimensional case the sums $\sigma(\vec{k})$ and $\sigma_c(\vec{k})$ have a series of peculiarities discussed in Refs. 31–33. Now we note only that the sum $\sigma(\vec{k})$ at $\vec{k} = 0$ converges for large $\vec{l}$ faster than in the three-dimensional case and has the finite value $\sigma(0) = 9.03362$. On the other hand, the representation $\sigma(\vec{k})$ through the integral applicable in the three-dimensional case (see Ref. 35) is not feasible since the corresponding two-dimensional integral $\int dx dy / (x^2 + y^2)^{3/2}$ diverges as $x, y \rightarrow 0$. Therefore, the regular presentation of the static demagnetization field $\vec{H}_m$ through the magnetization components $\vec{M}$ of the form $\vec{H}_m = -4\pi \sum_{i=1}^3 (\vec{e}_i N_i M_i)$, where N_i are demagnetizing factors, $\sum_{i=1}^3 N_i = 1$, fails in the two-dimensional case. In particular, Yafet and Georgy demonstrated³⁶ that for the atomic monolayer the z -projection of static demagnetizing field $\vec{H}_m$, is $(3/2)\sigma(0)$ in our notation, and it does not coincide with the continuum theory result for a thin film, $\vec{H}_m = -4\pi \vec{e}_z M_z$. The numerically obtained value of $\sigma(0) = 9.03362$ gives $(3/2)\sigma(0) = 1.0783 \cdot 4\pi$, thus the difference is nearly 8%. For $\vec{k} \neq 0$ the specific feature of the two-dimensional case manifests itself more vividly as the appearance of a non-analytical dependence on $\vec{k}$, namely, at $k \rightarrow 0$,

$$\sigma(\vec{k}) = \sigma(0) - k \cdot F(\vec{k}), \sigma_c(\vec{k}) = \frac{(k_x - ik_y)^2}{\sqrt{k_x^2 + k_y^2}} \cdot G(\vec{k}), \quad (8)$$

where $k = |\vec{k}|$, $G(\vec{k})$ and $F(\vec{k})$ are the real functions, analytic as $k \rightarrow 0$ with the limit values $F(\vec{k}) \rightarrow 2\pi a$, $G(\vec{k}) \rightarrow 2\pi a/3$ as $\vec{k} \rightarrow 0$ and it is invariant under symmetry transformations of the square lattice. Thus in the collective oscillation spectra non-analytic features appear which are absent in the case of a three-dimensional ferromagnet. From here it is seen that in contrast to $\sigma(\vec{k})$ the complex sum $\sigma_c(\vec{k})$ is not invariant with respect to symmetry transformations of the square lattice, but these transformations change only the phase factor, and therefore do not have an effect on the frequency. The frequency contains only $|\sigma_c(\vec{k})| =$